

Recap

- We have been considering a formalism for addressing the issue of generalization abilities of a learning algorithm.
- The general framework is that of empirical risk minimization.
- We are looking at the question of consistency of empirical risk minimization.
- Let us start by briefly reviewing the notation.

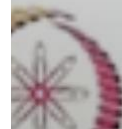


We are given

- $\mathcal{X}$ – input space; (as earlier, *Feature space*)
- $\mathcal{Y}$ – output space (as earlier, *Set of class labels*)
- $\mathcal{H}$ – hypothesis space (*family of classifiers*)

Each $h \in \mathcal{H}$ is a function, $h : \mathcal{X} \rightarrow \mathcal{A}$,
where $\mathcal{A}$ is called *action space*.

- Training data: $\{(X_i, y_i), i = 1, \dots, n\}$
drawn *iid* according to some distribution P_{xy} on
 $\mathcal{X} \times \mathcal{Y}$.



- Loss function: $L : \mathcal{Y} \times \mathcal{A} \rightarrow \mathbb{R}^+$.
- The risk function, $R : \mathcal{H} \rightarrow \mathbb{R}^+$, is given by

$$R(h) = E[L(y, h(X))] = \int L(y, h(X)) dP_{xy}$$

We assume that L is bounded so that the expectation always exists.

- Let $h^* = \arg \min_{h \in \mathcal{H}} R(h)$
- We define the goal of learning as finding h^* , the global minimizer of risk.



- However, we can not directly minimize $R(\cdot)$.
- The empirical risk function, $\hat{R}_n : \mathcal{H} \rightarrow \mathbb{R}^+$, is defined by

$$\hat{R}_n(h) = \frac{1}{n} \sum_{i=1}^n L(y_i, h(X_i))$$

- Let

$$\hat{h}_n^* = \arg \min_{h \in \mathcal{H}} \hat{R}_n(h)$$

- An algorithm learns $\hat{h}_n^*$ by minimizing empirical risk.



- Our objective is to find h^* , minimizer of risk $R(\cdot)$.
- Since we do not know R , we minimize $\hat{R}_n$ instead, and thus find $\hat{h}_n^*$.
- Hence we are interested in the question

$$R(\hat{h}_n^*) \rightarrow R(h^*)?$$

where the convergence is in probability.

- This is the issue of consistency of empirical risk minimization.



- As we saw earlier, if $\mathcal{H}$ is too 'flexible', minimizer of empirical risk does not necessarily have low risk.
- Suppose we use 0–1 loss function.
- Consider $h_1 : \mathcal{X} \rightarrow \mathcal{Y}$ with $h_1(X_i) = y_i$, $(X_i, y_i) \in S$ and $h_1(X) = 1$ for all other X
- Then $\hat{R}_n(h_1) = 0$! It is a global minimizer of empirical risk. But it is obvious that h_1 is not a good classifier.



- If functions like h_1 are in our $\mathcal{H}$ then empirical risk minimization (ERM) may not yield good classifiers.
- If $\mathcal{H}$ contains all possible functions, then this is certainly the case as we saw in our example.
- Functions like h_1 could be highly non-smooth and hence one way is to impose some smoothness conditions on the learnt function (e.g., regularization).
- Issue of consistency depends on $\mathcal{H}$, the class of functions over which we minimize empirical risk.
- We are trying to ask what $\mathcal{H}$ should satisfy so that ERM is consistent.



Consistency of Empirical Risk Minimization

- We would like the algorithm to satisfy: $\forall \epsilon, \delta > 0$, $\exists N < \infty$, such that

$$\text{Prob}[|R(\hat{h}_n^*) - R(h^*)| > \epsilon] \leq \delta, \forall n \geq N$$

- In addition, we would also like to have

$$\text{Prob}[|\hat{R}_n(\hat{h}_n^*) - R(h^*)| > \epsilon] \leq \delta, \forall n \geq N$$

We would like to (approximately) know the true risk of the learnt classifier.

- For what kind of $\mathcal{H}$ do these hold?



- As we already saw, the law of large numbers (that $\hat{R}_n(h) \rightarrow R(h), \forall h$) is not enough.
- As it turns out, what we need is that the convergence under law of large numbers be uniform over $\mathcal{H}$.
- Such uniform convergence is necessary and sufficient for consistency of empirical risk minimization.
- We now explain what is meant by uniform convergence.
- Law of large numbers says that sample mean converges to expectation of the random variable.



- In the context of empirical risk this means:
- Given any $h, \forall \epsilon, \delta > 0, \exists N < \infty$ such that

$$\text{Prob}[|\hat{R}_n(h) - R(h)| > \epsilon] \leq \delta, \forall n \geq N$$

- The N that exists can depend on ϵ, δ and **also on h** .
- The convergence is said to be uniform if the N depends only on ϵ, δ and not on h .
- That is, for a given ϵ, δ the same $N(\epsilon, \delta)$ works for all $h \in \mathcal{H}$.



- To sum up, $\hat{R}_n(h)$ converges (in probability) to $R(h)$ uniformly over $\mathcal{H}$ if $\forall \epsilon, \delta > 0, \exists N(\epsilon, \delta) < \infty$ such that

$$\text{Prob} \left[\sup_{h \in \mathcal{H}} |\hat{R}_n(h) - R(h)| > \epsilon \right] \leq \delta, \quad \forall n \geq N(\epsilon, \delta)$$

- This implies that the same $N(\epsilon, \delta)$ works for all $h \in \mathcal{H}$.
- It is easy to show that uniform convergence is sufficient for consistency of empirical risk minimization.



We have

$$\begin{aligned} R(\hat{h}_n^*) - R(h^*) &= [R(\hat{h}_n^*) - \hat{R}_n(\hat{h}_n^*)] + \\ &\quad [\hat{R}_n(\hat{h}_n^*) - \hat{R}_n(h^*)] + [\hat{R}_n(h^*) - R(h^*)] \\ &\leq [R(\hat{h}_n^*) - \hat{R}_n(\hat{h}_n^*)] + [\hat{R}_n(h^*) - R(h^*)] \end{aligned}$$

$(\hat{R}_n(\hat{h}_n^*) - \hat{R}_n(h^*)) \leq 0$ because $\hat{h}_n^*$ is minimizer of $\hat{R}_n$)

- Also, since h^* is minimizer of R ,

$$(R(\hat{h}_n^*) - R(h^*)) \geq 0.$$



- Hence we have

$$|R(\hat{h}_n^*) - R(h^*)| \leq |R(\hat{h}_n^*) - \hat{R}_n(\hat{h}_n^*)| + |\hat{R}_n(h^*) - R(h^*)|$$

- Because of uniform convergence, we can make both terms on the RHS less than $\epsilon/2$, with a high probability, for large n and hence can make the LHS less than ϵ with a large probability.
- This shows consistency of ERM.
- Since arguments like this are needed many times here, let us argue the above more precisely.



- Because of uniform convergence,
 $\forall \epsilon, \delta > 0, \exists N(\epsilon, \delta) < \infty$, **s.t.** $\forall n \geq N(\epsilon, \delta)$,

$$\text{Prob} \left[|R(\hat{h}_n^*) - \hat{R}_n(\hat{h}_n^*)| > \frac{\epsilon}{2} \right] \leq \frac{\delta}{2}, \quad \text{and}$$

$$\text{Prob} \left[|\hat{R}_n(h^*) - R(h^*)| > \frac{\epsilon}{2} \right] \leq \frac{\delta}{2}$$

- Using this, we have to show

$$\text{Prob}[|R(\hat{h}_n^*) - R(h^*)| > \epsilon] \leq \delta, \quad \forall n \geq N(\epsilon, \delta)$$



- Define events A, B, C by

$$A = [|R(\hat{h}_n^*) - R(h^*)| \leq \epsilon], \quad B = [|R(\hat{h}_n^*) - \hat{R}_n(\hat{h}_n^*)| \leq \frac{\epsilon}{2}],$$

$$C = [|\hat{R}_n(h^*) - R(h^*)| \leq \frac{\epsilon}{2}]$$

- Since

$$|R(\hat{h}_n^*) - R(h^*)| \leq |R(\hat{h}_n^*) - \hat{R}_n(\hat{h}_n^*)| + |\hat{R}_n(h^*) - R(h^*)|,$$

we have $A \supset (B \cap C)$ and hence $A^c \subset (B^c \cup C^c)$



- This gives us

$$\text{Prob}[A^c] \leq \text{Prob}[B^c \cup C^c] \leq \text{Prob}[B^c] + \text{Prob}[C^c]$$

- By uniform convergence, probability of both B^c and C^c are less than $\delta/2$. Hence,

$$\text{Prob}[A^c] = \text{Prob}[|R(\hat{h}_n^*) - R(h^*)| > \epsilon] \leq \delta$$

- That shown consistency of empirical risk minimization.



Consistency of Empirical Risk Minimization

- For consistency, the algorithm should satisfy:
 $\forall \epsilon, \delta > 0, \exists N < \infty$, such that

$$\text{Prob}[|R(\hat{h}_n^*) - R(h^*)| > \epsilon] \leq \delta, \forall n \geq N$$

- We have shown this. In addition, we wanted

$$\text{Prob}[|\hat{R}_n(\hat{h}_n^*) - R(h^*)| > \epsilon] \leq \delta, \forall n \geq N$$



- We can show this also as follows (using the uniform convergence)



- We have

$$|\hat{R}_n(\hat{h}_n^*) - R(h^*)| \leq \underbrace{|\hat{R}_n(\hat{h}_n^*) - R(\hat{h}_n^*)|}_{\text{by triangular inequality}} + |R(\hat{h}_n^*) - R(h^*)|$$

by triangular inequality.

- By uniform convergence, for sufficiently large n , we can make both terms on the RHS smaller than $\epsilon/2$ with a large probability. Hence we can make the LHS smaller than ϵ with a large probability (for large n).
- This gives us the result we want.



- Thus convergence of $\hat{R}_n(h)$ to $R(h)$ uniformly over $\mathcal{H}$ is sufficient for consistency of empirical risk minimization.
- This uniform convergence is also necessary for the consistency.
- The proof of necessity is much more involved and hence we will not present it in this course.



- The next question is, given a $\mathcal{H}$, how do we know whether the needed uniform convergence holds.
- We need some useful characterization of family of functions for which this uniform convergence holds.
- That is what we do next.
- For simplicity we consider only family of binary-valued functions on $\mathcal{X}$. (That is, we are considering 2-class problems with $\mathcal{Y} = \mathcal{A} = \{0, 1\}$).
- We also assume that $L(y, h(X)) \in [0, 1]$.



- First we note that if $\mathcal{H}$ is finite then the uniform convergence always holds.
- Suppose $\mathcal{H} = \{h_1, h_2, \dots, h_M\}$.
- By law of large numbers, for any h_i , given any $\epsilon, \delta > 0$, there would be a $N_i(\epsilon, \delta)$ such that

$$\text{Prob}[|\hat{R}_n(h_i) - R(h_i)| > \epsilon] \leq \delta, \quad \forall n > N_i(\epsilon, \delta)$$

- Take $N(\epsilon, \delta) = \max_i N_i(\epsilon, \delta)$.
- This N would work for all h_i and hence we have uniform convergence.



- Let us actually calculate the bound on the examples needed, namely, N .
- For this we try to bound $\text{Prob}[|\hat{R}_n(h_i) - R(h_i)| > \epsilon]$ with a function of n .
- If we want to use, e.g., Chebyshev inequality for this, we need moments of random variables $L(y, h_i(X))$. But we may not have such information.
- Hence we would use some distribution independent bounds for this.



- Let Z_i be *iid* random variables taking values in $[a, b]$, with mean μ . Then the two sided Hoeffding inequality is

$$\text{Prob} \left[\left| \frac{1}{n} \sum_{i=1}^n Z_i - \mu \right| > \epsilon \right] \leq 2 \exp \left(-\frac{2n\epsilon^2}{(b-a)^2} \right)$$

- This gives a distribution independent bound and hence we can use this.



- Take $Z_i = L(y_i, h(X_i))$. Then Z_i are *iid* random variables taking values in $[0, 1]$.
- Then $\frac{1}{n} \sum Z_i = \hat{R}_n(h)$ and $EZ_i = R(h)$.
- Hence, for any h , we have

$$\text{Prob} \left[|\hat{R}_n(h) - R(h)| > \epsilon \right] \leq 2 \exp(-2n\epsilon^2)$$



- Recall that $\mathcal{H} = \{h_1, \dots, h_M\}$.
- In the probability space corresponding to drawing n *iid* samples according to P_{xy} , define the events

$$C_\epsilon^i = \left[|\hat{R}_n(h_i) - R(h_i)| > \epsilon \right], \quad i = 1, \dots, M$$

- We have just seen that

$$\text{Prob}(C_\epsilon^i) \leq 2 \exp(-2n\epsilon^2), \quad \forall i$$



Now we have

$$\begin{aligned}\text{Prob} \left[\sup_{h \in \mathcal{H}} |\hat{R}_n(h) - R(h)| > \epsilon \right] &= \text{Prob} (C_\epsilon^1 \cup \dots \cup C_\epsilon^M) \\ &\leq \sum_{i=1}^M \text{Prob}(C_\epsilon^i) \\ &\leq 2M \exp(-2n\epsilon^2)\end{aligned}$$

- Now we can find how large n should be so that the bound on the RHS is less than δ



- One situation where we can take $\mathcal{H}$ to be finite is when we have Boolean features.
- Suppose we have d Boolean features. Then $\mathcal{X}$ is set of all d -bit Boolean numbers and $2^{\mathcal{X}}$ is a finite set.
- In such cases, we know that ERM is always consistent. However, taking $\mathcal{H} = 2^{\mathcal{X}}$ would still not be nice.
- Here, $\mathcal{X}$ itself is finite with 2^d elements. Hence What is important is the sample complexity.



- We have shown

$$\text{Prob} \left[\sup_{h \in \mathcal{H}} |\hat{R}_n(h) - R(h)| > \epsilon \right] \leq 2M \exp(-2n\epsilon^2)$$

- We can bound this probability by δ if

$$2M \exp(-2n\epsilon^2) \leq \delta \quad \text{or} \quad n \geq \frac{1}{2\epsilon^2} \ln \left(\frac{2M}{\delta} \right)$$

- The bound may be poor (and may be more than 2^d).



- For specific finite $\mathcal{H}$ we can get better bounds.
- There are classes of Boolean functions that can be learnt efficiently.
- But, for us, the reason for doing the finite $\mathcal{H}$ case is that it gives us ideas on how to tackle the general case.



- Now let $\mathcal{H}$ be arbitrary.
- Given any h , the value of $\hat{R}_n(h)$ is calculated based on n iid samples.
- Given $h, h' \in \mathcal{H}$, if $h(X_i) = h'(X_i)$, $i = 1, \dots, n$, then, $\hat{R}_n(h) = \hat{R}_n(h')$.
- Since each h is a binary valued function, on the n training samples, X_i , there are only 2^n tuples of distinct values any function can take.
- Hence based on the values of $\hat{R}_n(h)$ we can only distinguish finitely many functions from $\mathcal{H}$.



- We can sum-up this insight as follows.
- Given n training examples, as far as empirical risk is concerned, only finitely many (at most 2^n) functions from $\mathcal{H}$ can be distinguished.
- Hence we may be able to employ the argument we used for finite $\mathcal{H}$ case to tackle the general case.



- Recall that in the finite case we had

$$\text{Prob} \left[\sup_{h \in \mathcal{H}} |\hat{R}_n(h) - R(h)| > \epsilon \right] \leq 2M \exp(-2n\epsilon^2)$$

- Using our insight we may be able to use this but then M would be a function of n . So, this depends on how the number of distinguishable functions grow with n , the number of examples.
- Next, we explore this intuitive idea in a more precise fashion.

